run_sjsdm_cv_calibration_benchmark.R

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`run_sjsdm_cv_calibration_benchmark`
Run an Adaptive sjSDM CV Budget Calibration

Description

Screens all candidates on three folds at a reduced sampling budget. After each stable adjacent-rung pair, it confirms the winner and runner-up on all five folds of repeats one and two at the production sampling budget.

Usage

```
run_sjsdm_cv_calibration_benchmark(
  data_candidates = NULL,
  list_prepared_folds = NULL,
  sel_abiotic_formula = NULL,
  config_sjsdm_cv_fitting = NULL,
  data_ladder = build_sjsdm_cv_calibration_ladder(),
  screening_fold_ids = 1:3,
  confirmation_fold_ids = 1:5,
  confirmation_sampling = 200L,
  n_confirmation_candidates = 2L,
  repeat_confirmation_loss_tolerance = 0.02,
  checkpoint_file = NULL,
  maximum_budget_order = Inf,
  rung_function = run_sjsdm_cv_calibration_rung,
  verbose = TRUE,
  ...
)
```

Arguments

`data_candidates`, `list_prepared_folds`, `sel_abiotic_formula`	Calibration model inputs.
`config_sjsdm_cv_fitting`	Validated structural CV fitting configuration.
`data_ladder`	Ordered budget ladder from [build_sjsdm_cv_calibration_ladder()].
`screening_fold_ids`, `confirmation_fold_ids`	Deterministic fold IDs used for screening and confirmation.
`confirmation_sampling`	Production sampling budget used for confirmation.
`n_confirmation_candidates`	Number of leading screening candidates retained for confirmation.
`repeat_confirmation_loss_tolerance`	Maximum relative loss gap allowed above the repeat-two winner.
`checkpoint_file`	Optional '.qs' path used to resume completed rungs after interruption.
`maximum_budget_order`	Optional execution bound. Stops after this ladder rung without changing the ladder signature or scientific acceptance gates. Defaults to 'Inf'.
`rung_function`	Injectable rung executor for deterministic orchestration tests.
`verbose`	Logical. If 'TRUE', progress is printed to the console.
`...`	Additional arguments passed to [run_sjsdm_cv_calibration_rung()].

Value

Calibration status, accepted budget when available, benchmark evidence, and attempt-level provenance accumulated before stopping.

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